🌐💻 IPv4 Addressing Part 1 (10.24.2026)

🎯 Big Picture

Moving from **Layer 2 (Switching/MAC)** to **Layer 3 (Routing/IP)**.
Layer 3 handles logical addressing and path selection to allow communication between *different* networks.

🧱 IPv4 Basics

- **32-bit logical address**, split into 4 sections called **Octets** (8 bits each).
- Written in **Dotted Decimal Notation** for human readability (e.g., `192.168.1.1`).
- **Binary (Base 2):** Computers read IPs in 1s and 0s.
- **Decimal (Base 10):** Humans read IPs in numbers 0-255.
- **Octet Value:** An 8-bit octet can hold a maximum value of **255** (all 1s: `11111111`).

🔀 Network Portion vs. Host Portion

Every IP address is split into two parts:

1. **Network Portion:** Identifies the specific network. (Like a street name).
2. **Host Portion:** Identifies the specific device on that network. (Like a house number).

How do we know the split?

- **Prefix Length:** Shorthand for the number of network bits (e.g., `/24` means 24 bits are for the network).
- **Netmask (Subnet Mask):** Dotted decimal representation of the prefix (e.g., `/24` = `255.255.255.0`).

> **Example:** `192.168.1.10 /24`
>
> - Network portion: `192.168.1`
> - Host portion: `.10`

🏷️ IPv4 Address Classes (Classful)

(See your cheat sheet image for ranges)

- **Class A:** `0-127` (Default /8) → Massive networks
- **Class B:** `128-191` (Default /16) → Medium networks
- **Class C:** `192-223` (Default /24) → Small networks
- *Note:* Modern networks use CIDR (Classless), but you must memorize these defaults for the CCNA.

📍 Special Addresses

- **Network Address:** The very first address in a network (all host bits = `0`). *Cannot be assigned to a device.*
  - Example: `192.168.1.0 /24`
- **Broadcast Address:** The very last address in a network (all host bits = `1`). *Sends data to all devices in the network.*
  - Example: `192.168.1.255 /24`
- **Loopback Address:** `127.x.x.x` (usually `127.0.0.1`). Used to test the local device's TCP/IP stack (localhost).
- **Multicast Address (Class D):** `224.0.0.0` to `239.255.255.255`. *Used for one-to-many communication. A single packet sent to a specific group of devices that have joined that group.*
  - Example: `224.0.0.5` (OSPF routers)
- **Experimental Address (Class E):** `240.0.0.0` to `255.255.255.255`. *Reserved by the IETF for experimental or future use. Non-routable and typically dropped by modern devices.*
  - Example: `240.0.0.1`

🧪 Practice Quiz: Binary & Decimal Conversions

Binary → Dotted Decimal

Tip: Remember the place values: `128 | 64 | 32 | 16 | 8 | 4 | 2 | 1`

- `00111111 00111000 11100111 00010011` = 63.56.231.19
- `11110011 01111111 01100010 00000001` = 243.127.98.1

- `01101111 00000110 01011001 11000111` = 111.6.89.199
- `11001111 11000110 00101111 01001100` = 207.198.47.76
- `01100100 11001001 00100001 11111101` = 100.201.33.253

Decimal → Binary

- `88.46.90.91` = `01011000 | 00101110 | 01011010 | 01011011`
- `221.234.246.163` = `11011101 | 11101010 | 11110110 | 10100011` *(Fixed last octet)*
- `3.41.143.222` = `00000011 | 00101001 | 10001111 | 11011110` *(Added)*
- `10.200.231.91` = `00001010 | 11001000 | 11100111 | 01011011` *(Added)*
- `248.87.255.152` = `11111000 | 01010111 | 11111111 | 10011000` *(Added)*

🧠 Quick Memory Hooks

- **First Octet Class Check:**
  - **A** = 1-126
  - **B** = 128-191
  - **C** = 192-223
- **Host Math:** Total usable hosts in a subnet = `2^(host bits) - 2` (Subtract 2 for the Network and Broadcast addresses).
- **The "Magic Number":** To quickly find the network address, subtract the subnet mask octet from 256 (e.g., Mask `255.255.255.192` → `256 - 192 = 64`. The network addresses go in increments of 64: `.0, .64, .128, .192`).